Engineering Curves: Construction of Cycloid

Rolling Circle Method with Tangent & Normal

Gajavada Sanjeevkumar

Problem Statement

Question

Draw a **cycloid** generated by a circle of diameter $D = 50$ mm rolling along a straight base line for **one complete revolution** without slipping.

Construct a **tangent** and a **normal** to the cycloid at any point P located on the curve.

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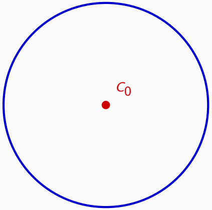

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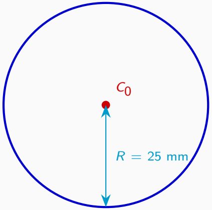

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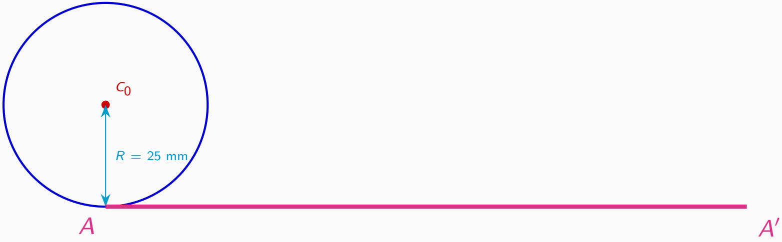

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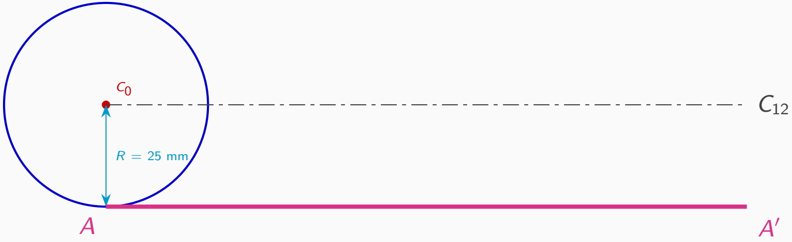

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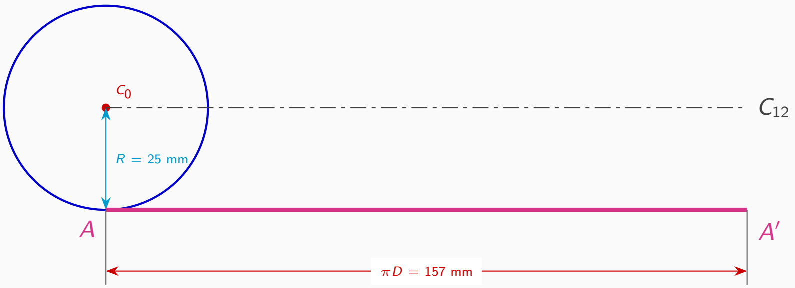

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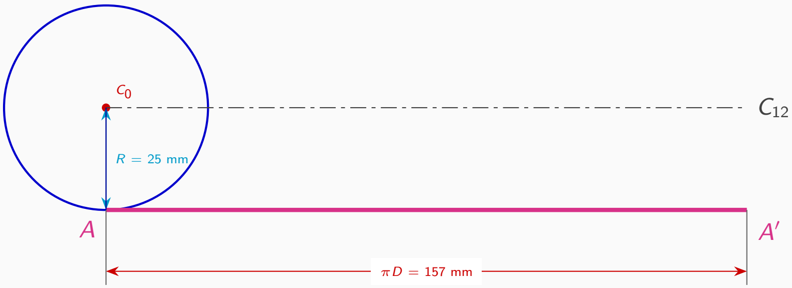

Step 2: Divide Generating Circle into 12 Equal Parts

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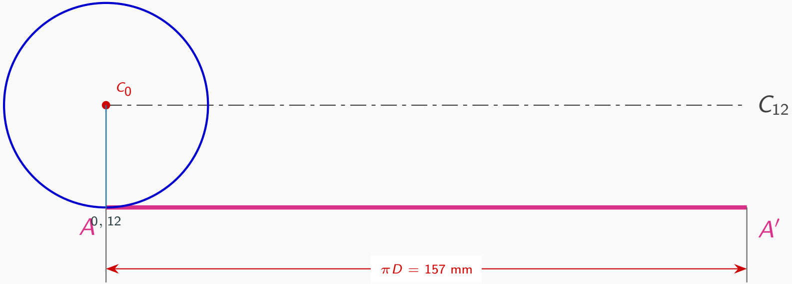

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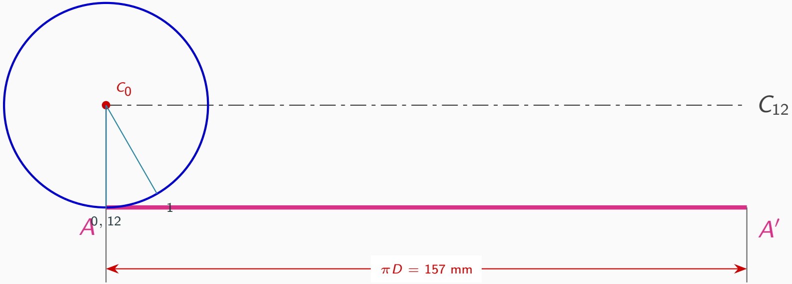

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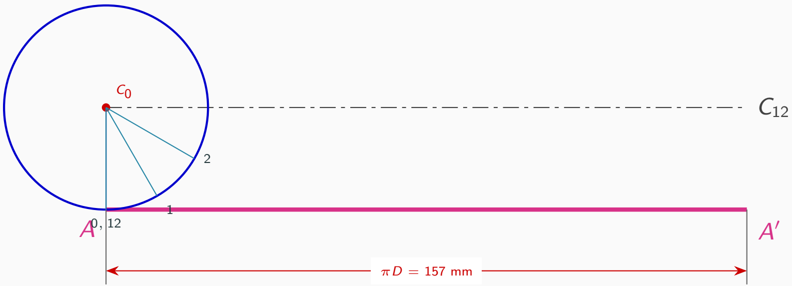

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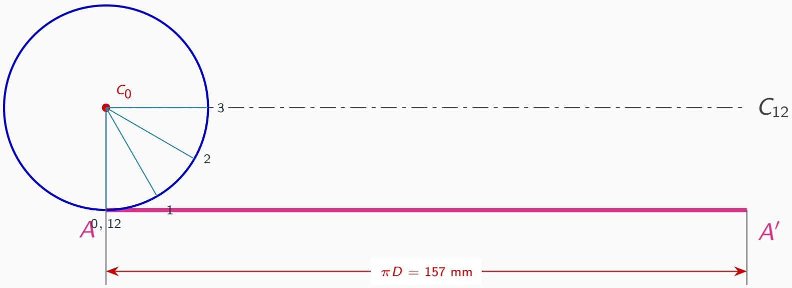

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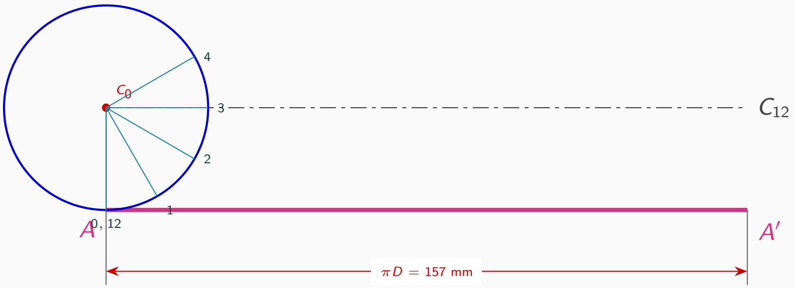

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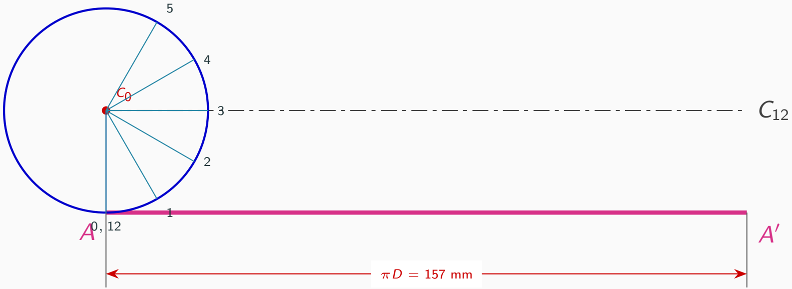

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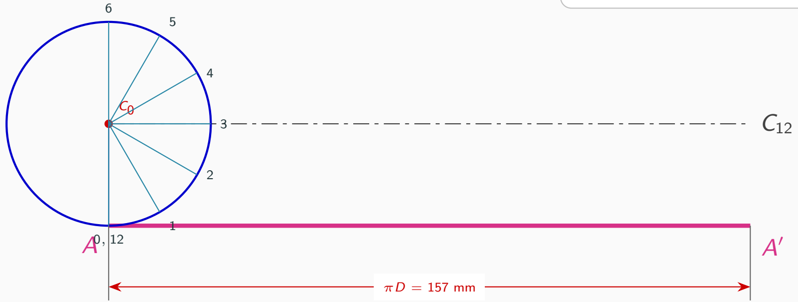

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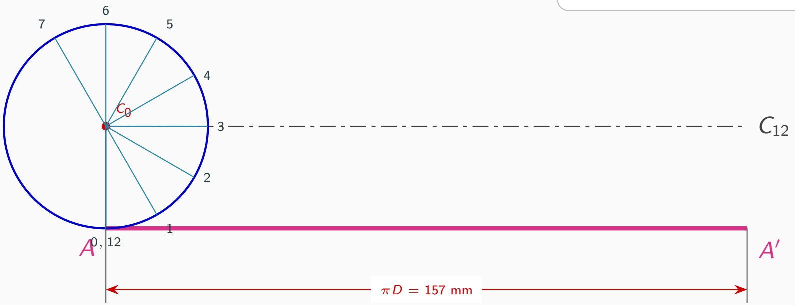

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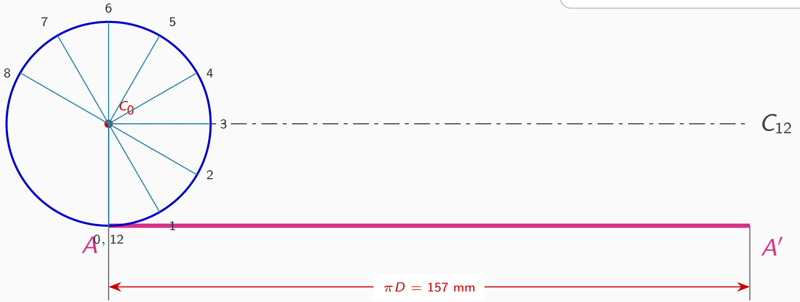

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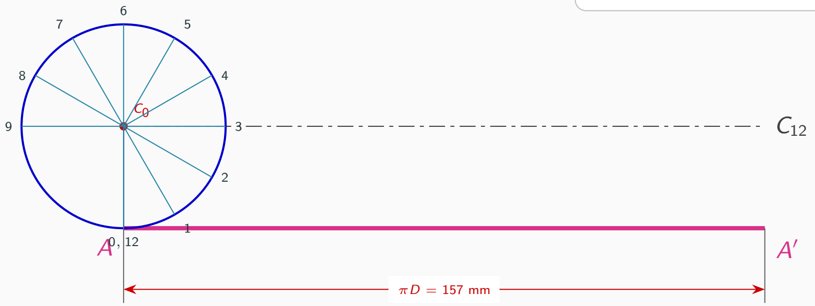

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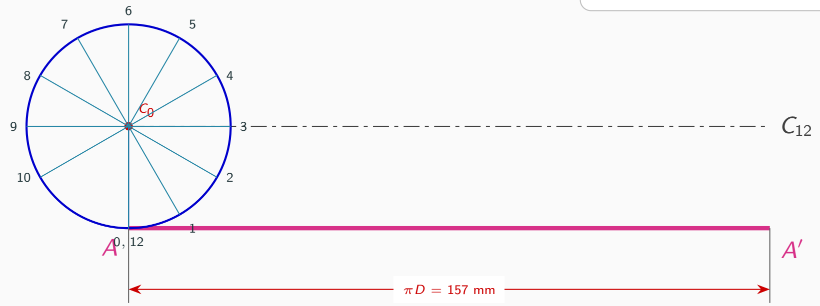

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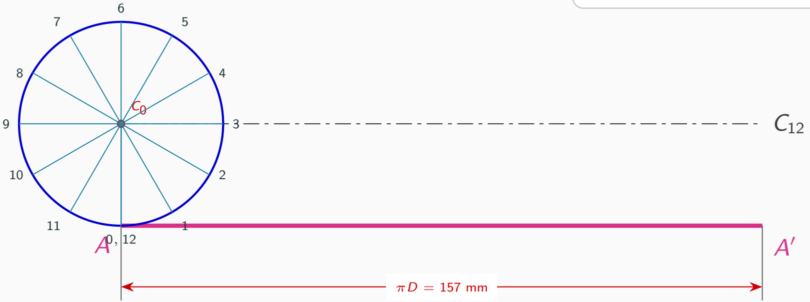

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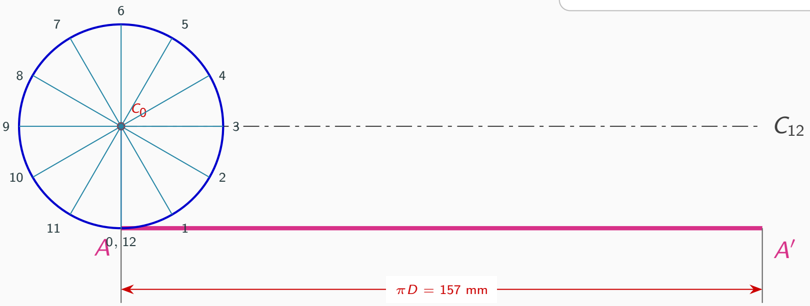

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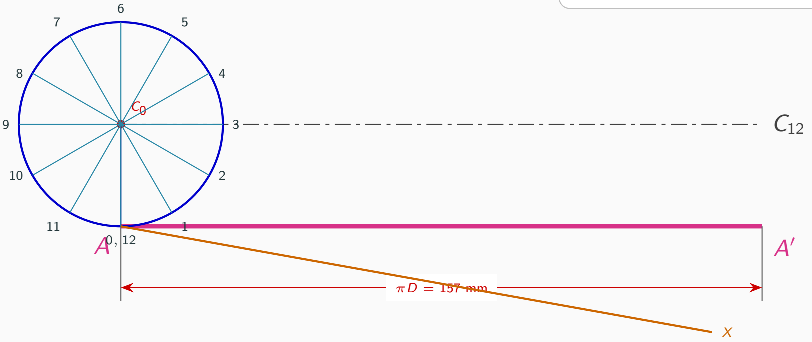

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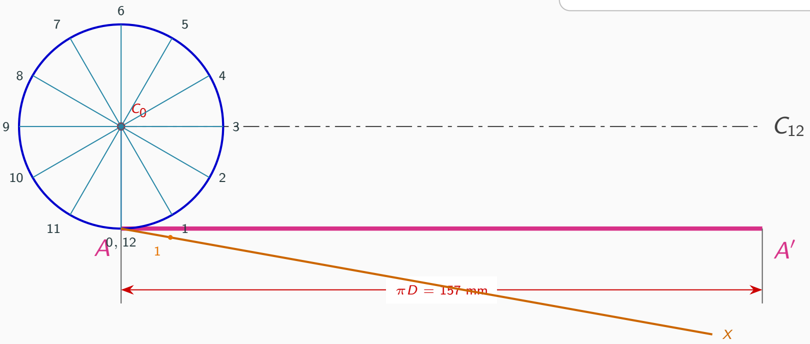

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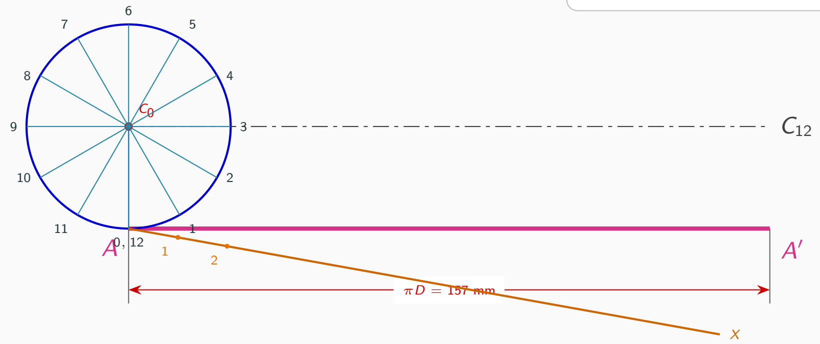

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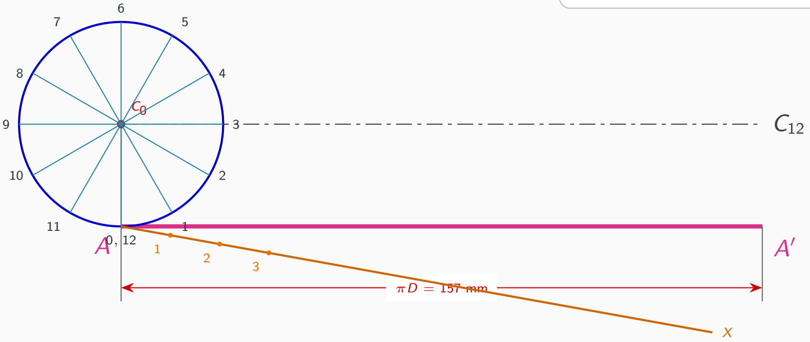

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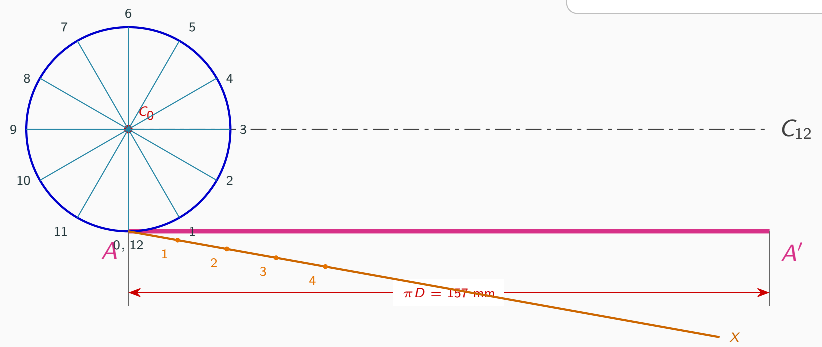

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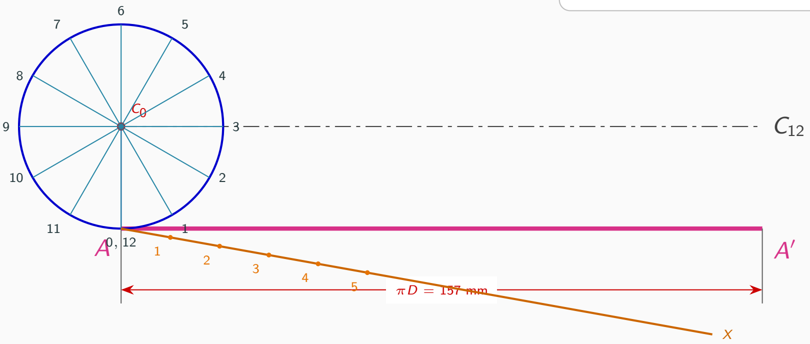

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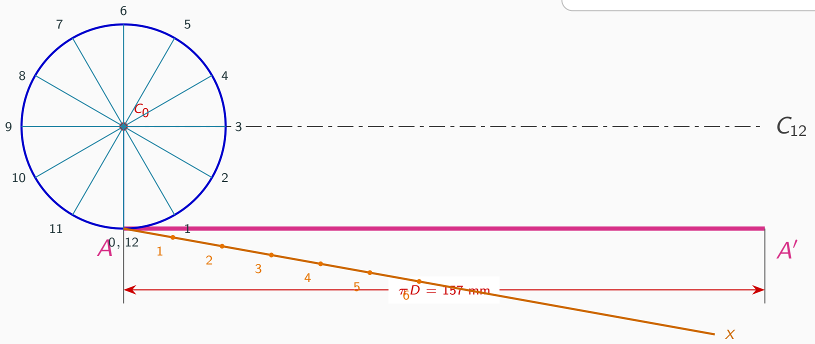

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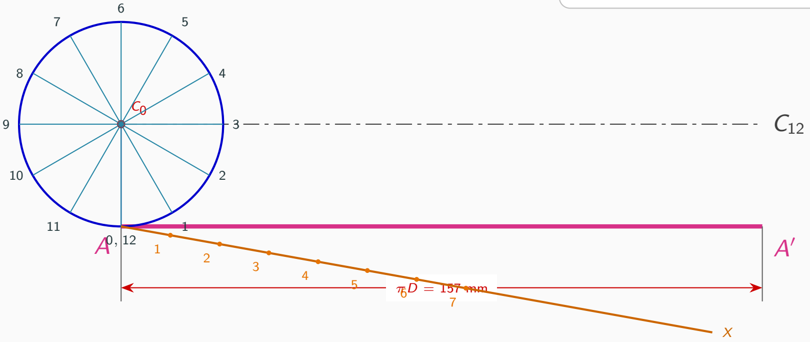

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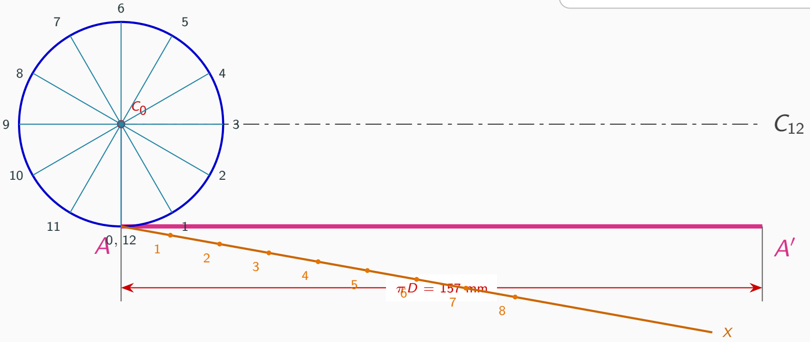

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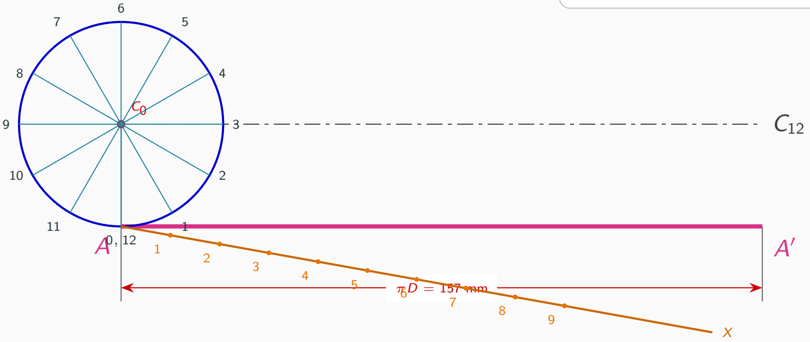

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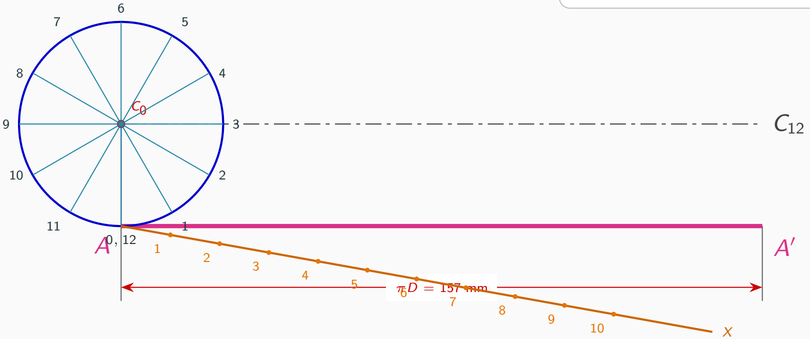

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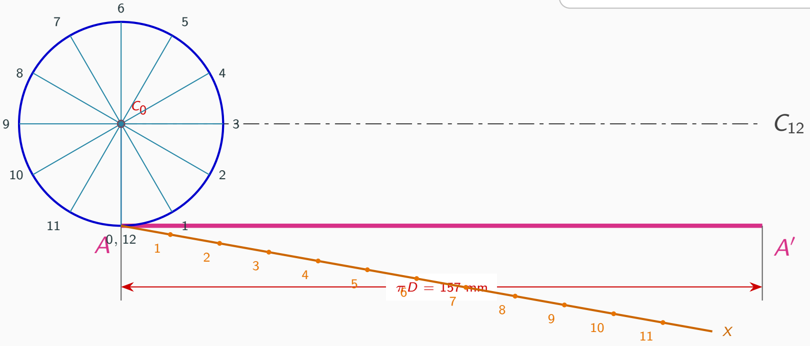

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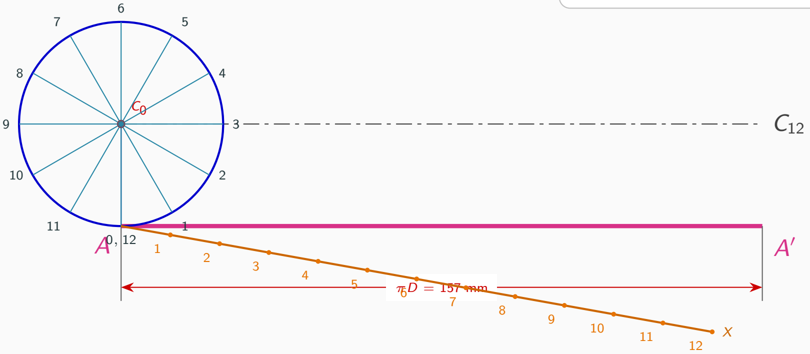

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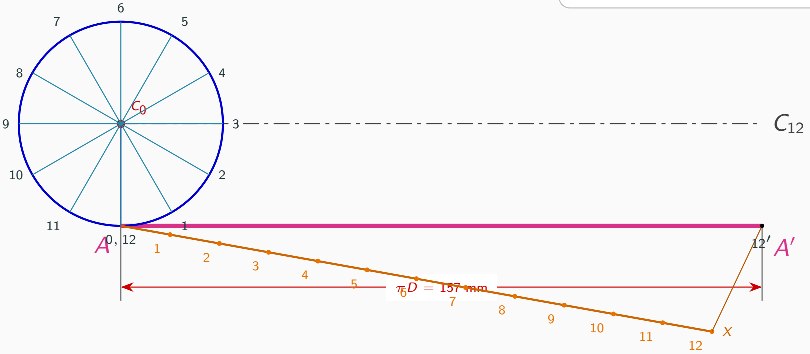

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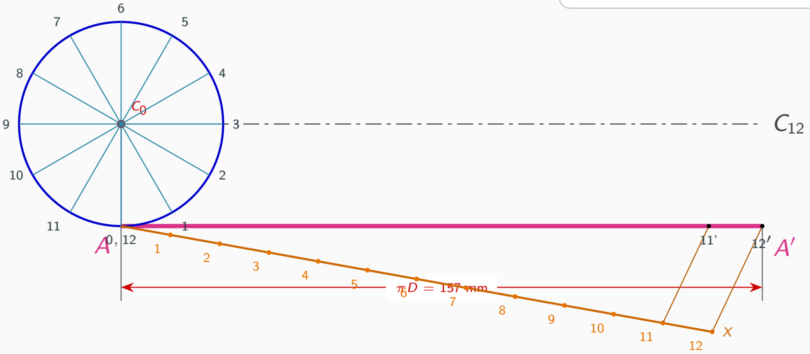

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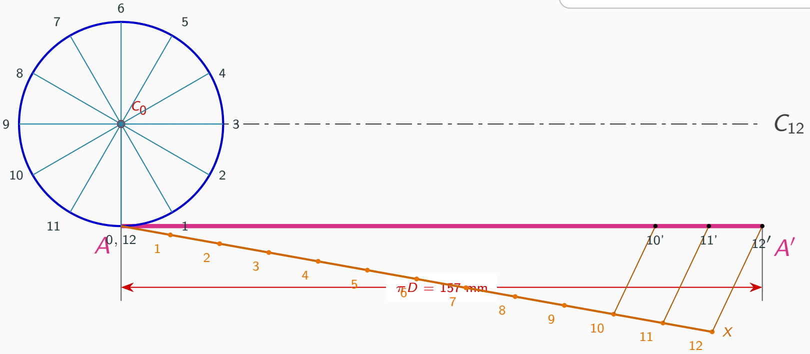

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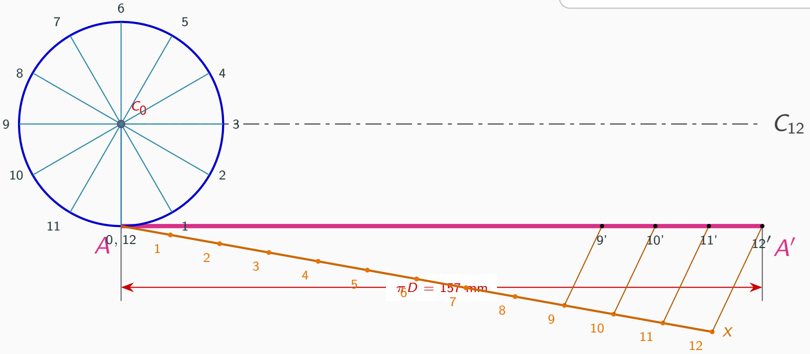

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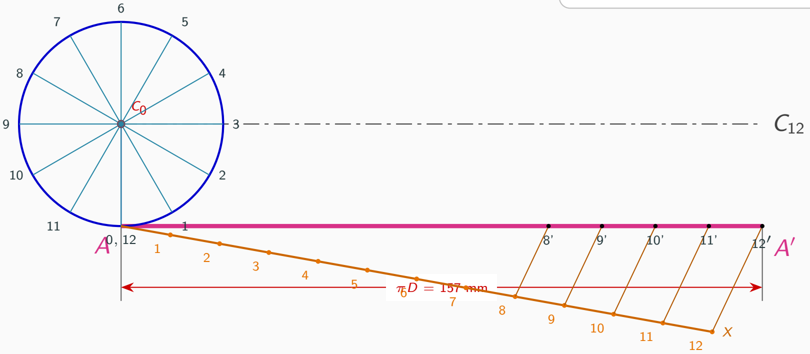

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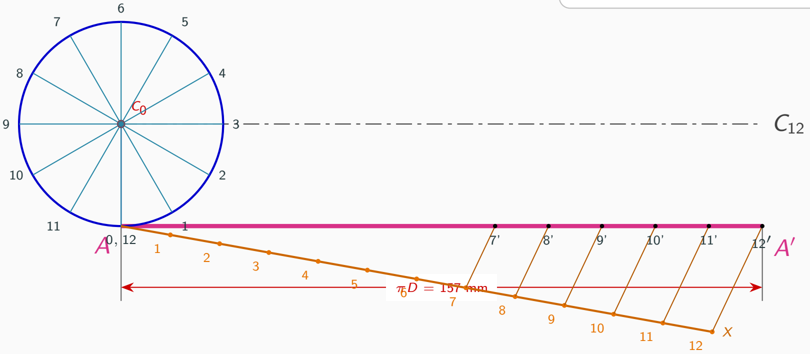

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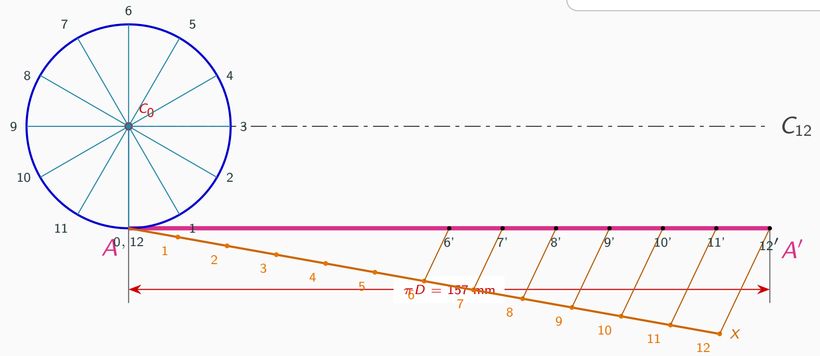

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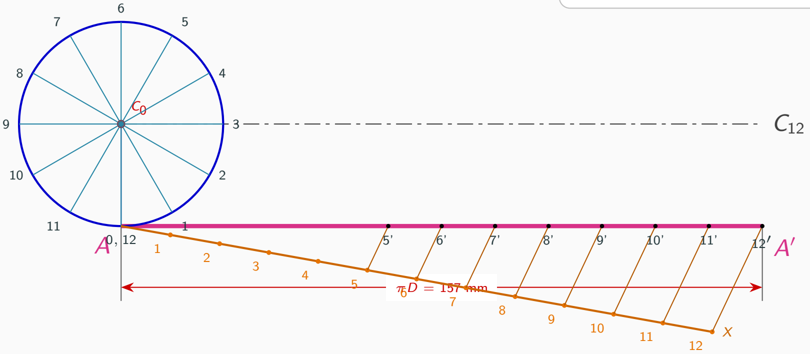

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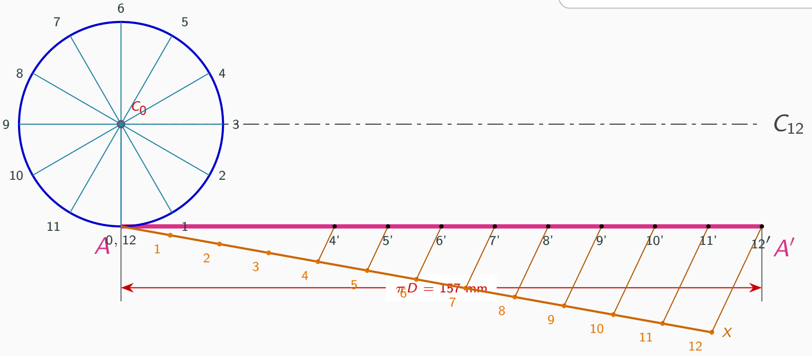

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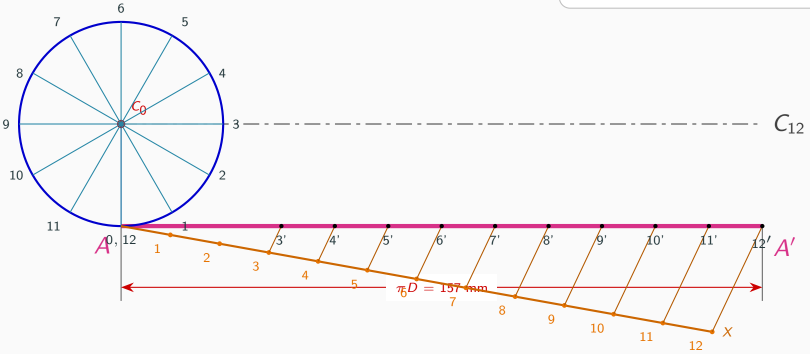

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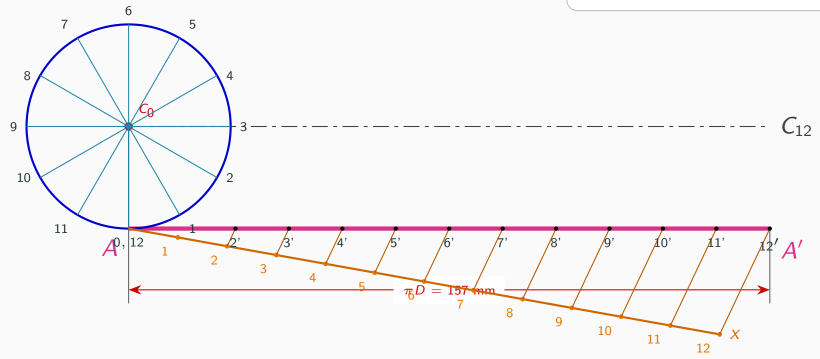

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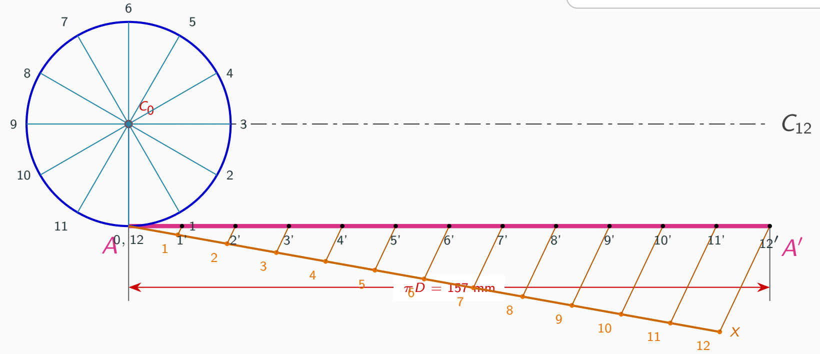

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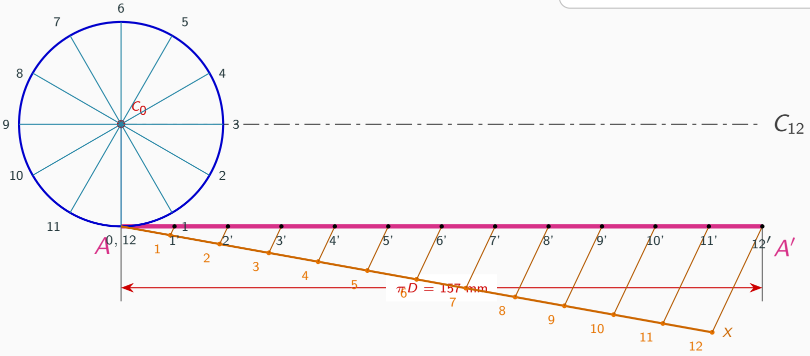

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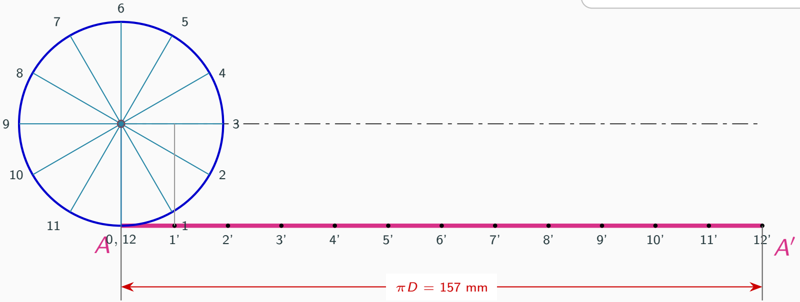

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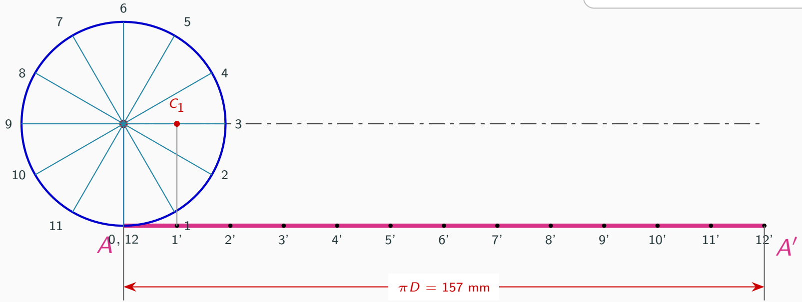

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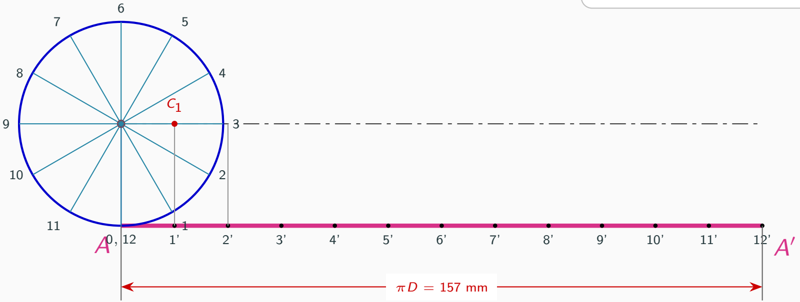

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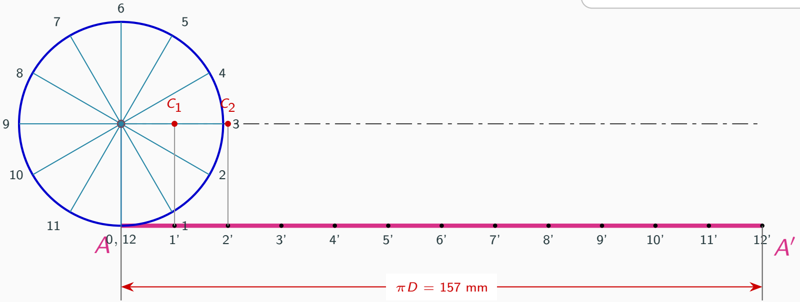

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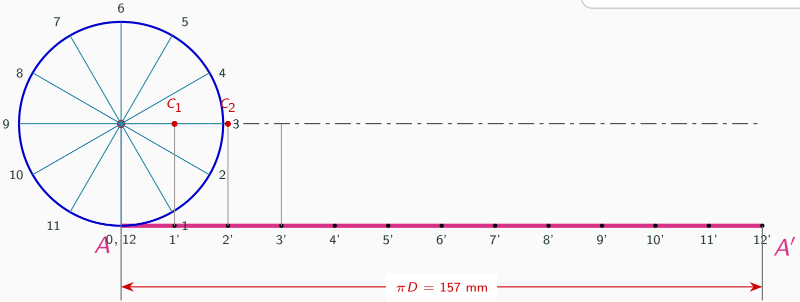

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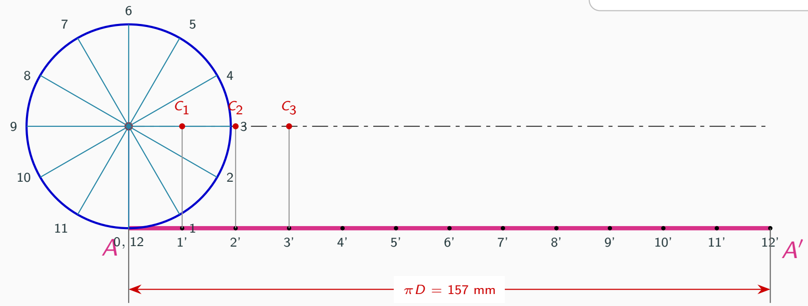
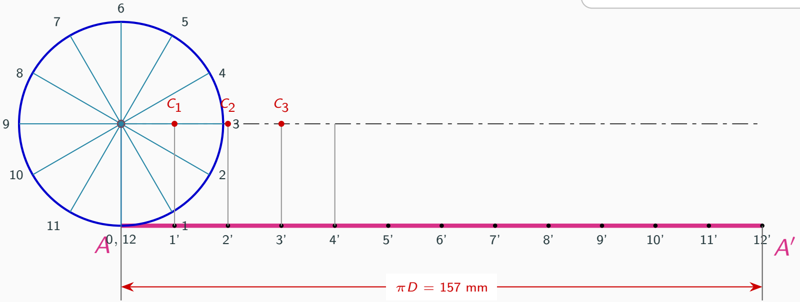


Step 4: Project Divisions to Locate Centers $C_1 \ldots C_{12}$

AA' $\rightarrow$ Directing Line
Axis Line $\rightarrow$ Center Line
$C_1 \ldots C_{12}$ $\rightarrow$ Center Points



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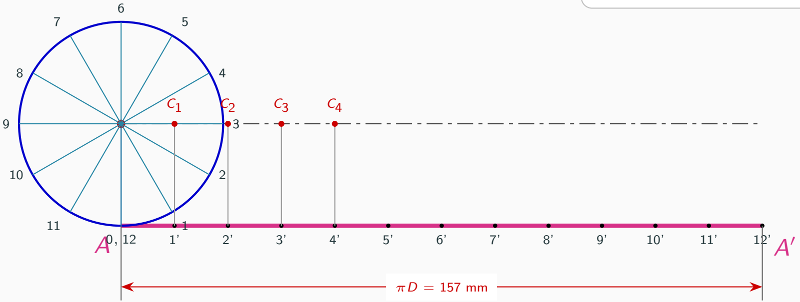

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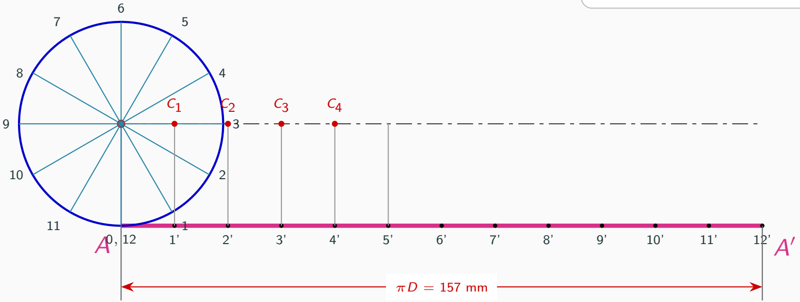

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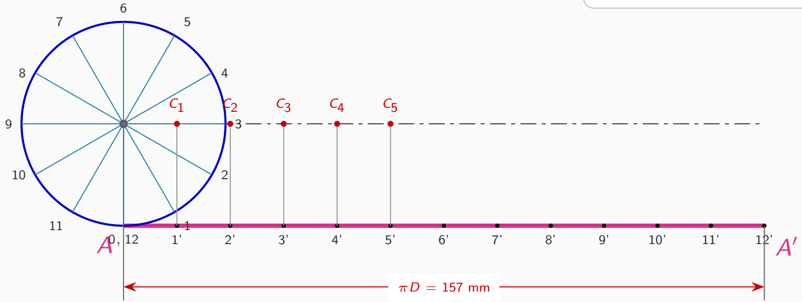

AA' → Directing Line
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$C_1 \dots C_{12}$ → Center Points

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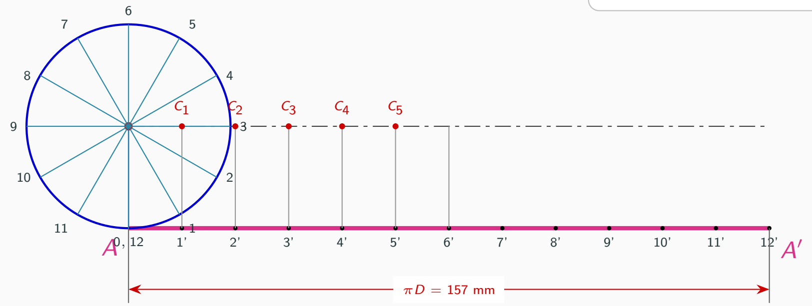

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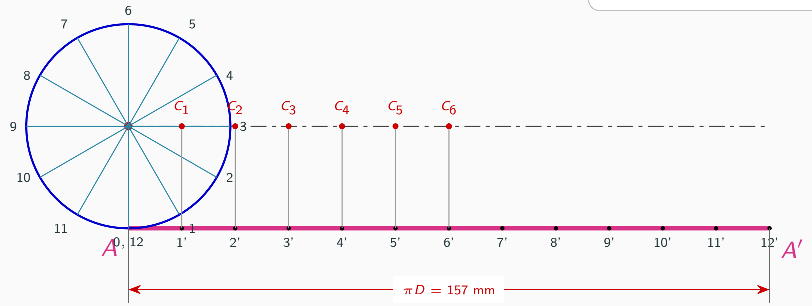

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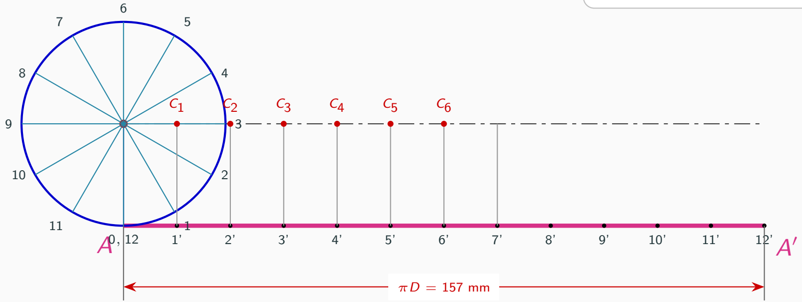

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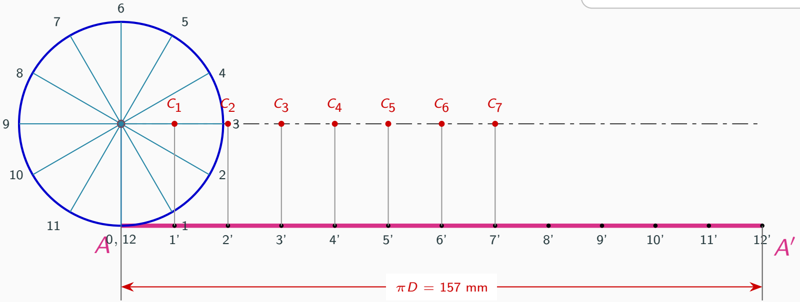

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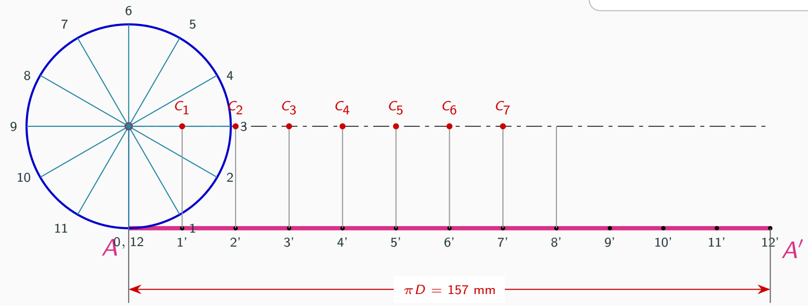

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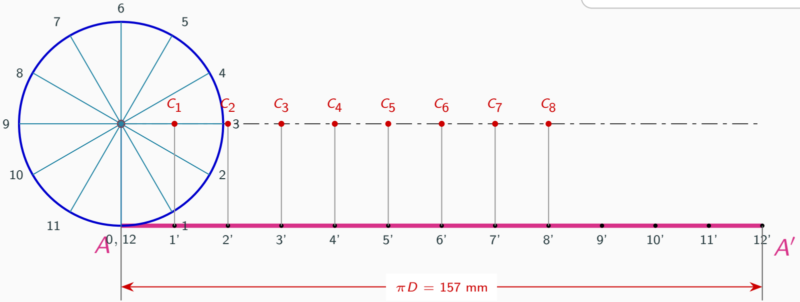

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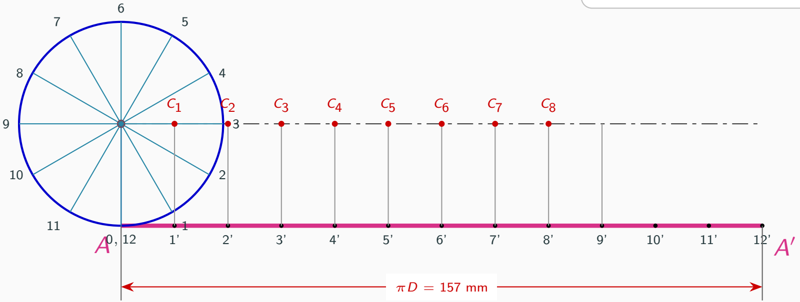

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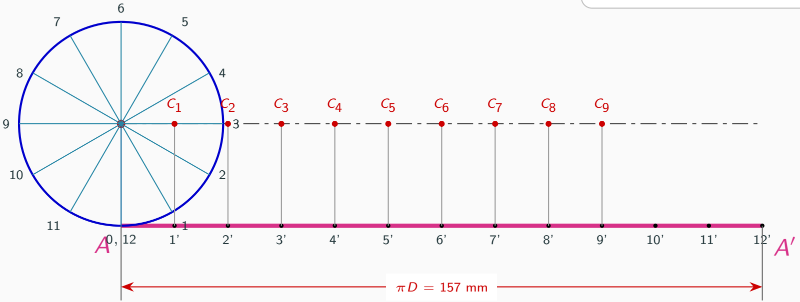

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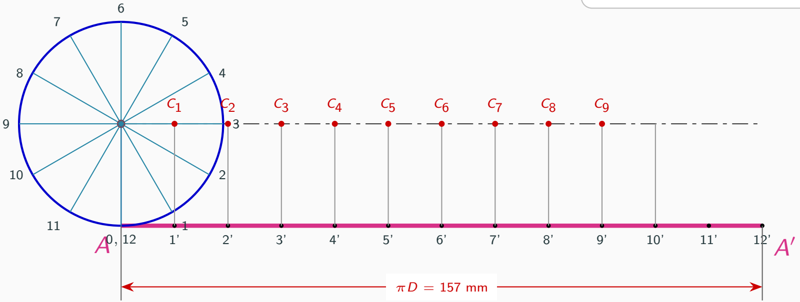

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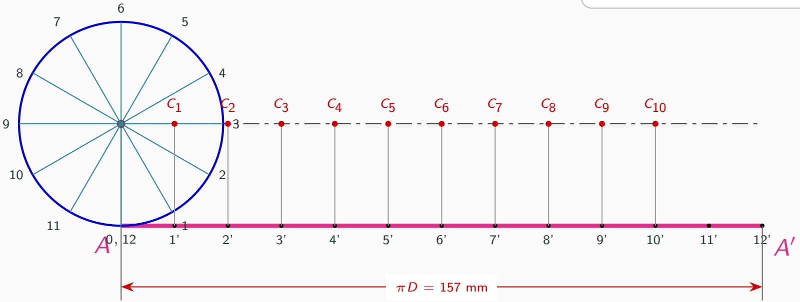

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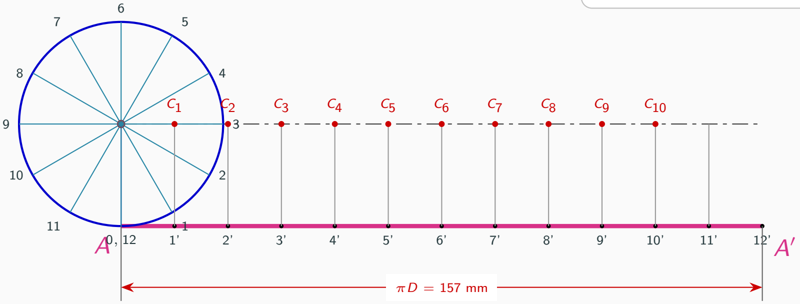

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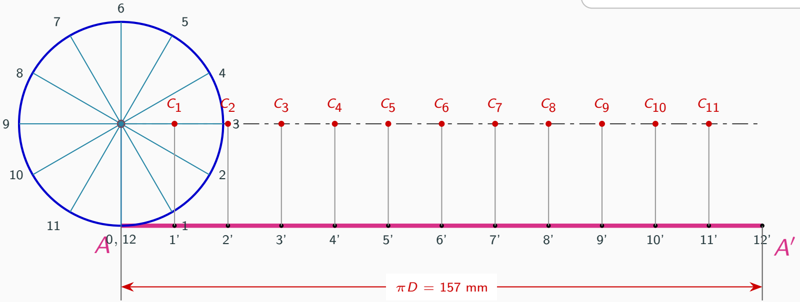

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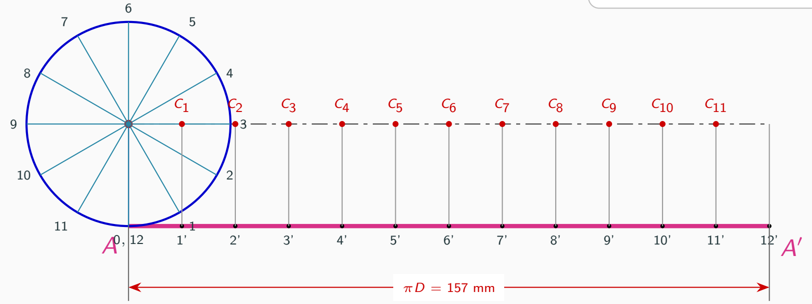

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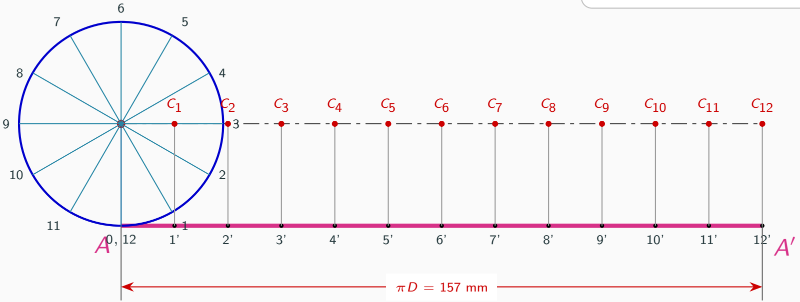

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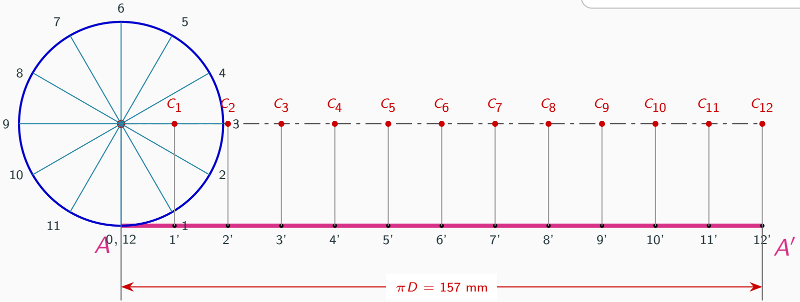

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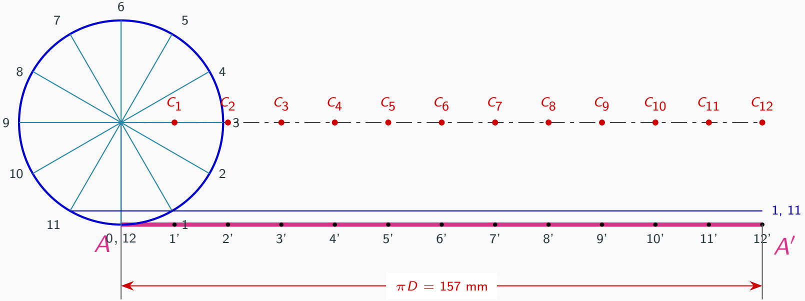
Step 5: Draw Horizontal Reference Lines

Blue Lines $\longrightarrow$ Horizontal Reference Lines
$C_1 \ldots C_{12}$ $\longrightarrow$ Retained Centers



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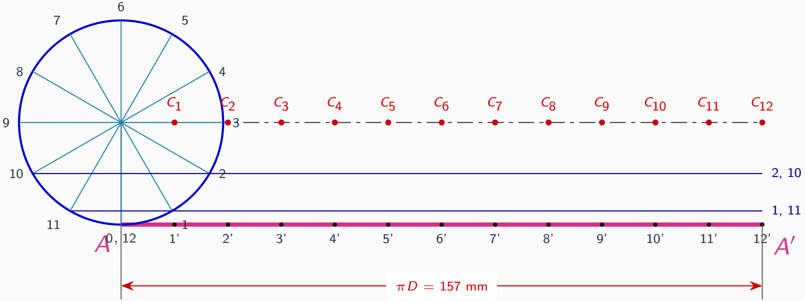

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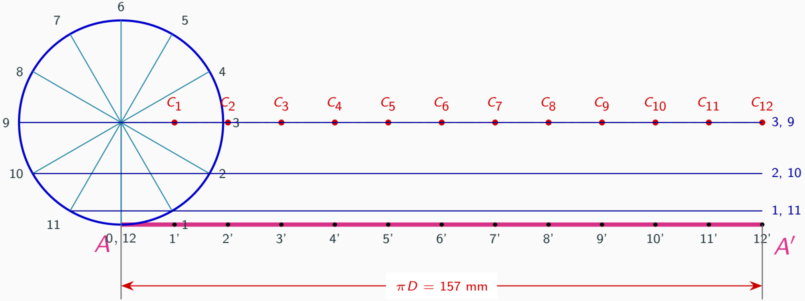

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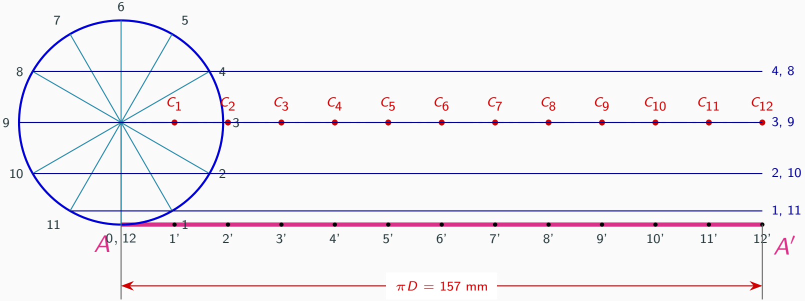

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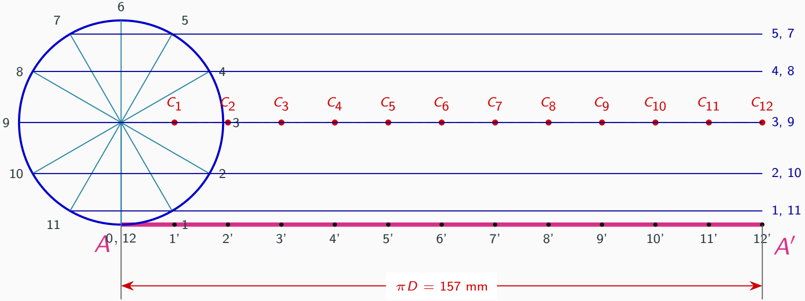

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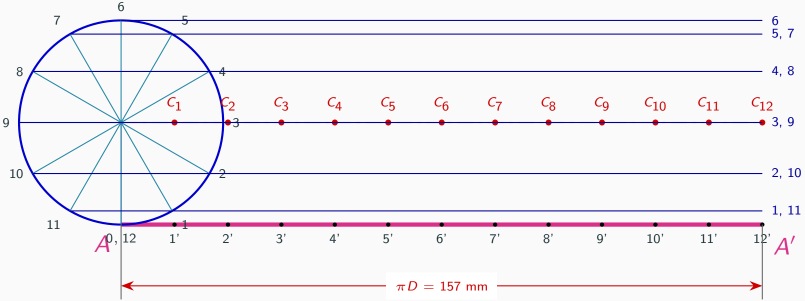
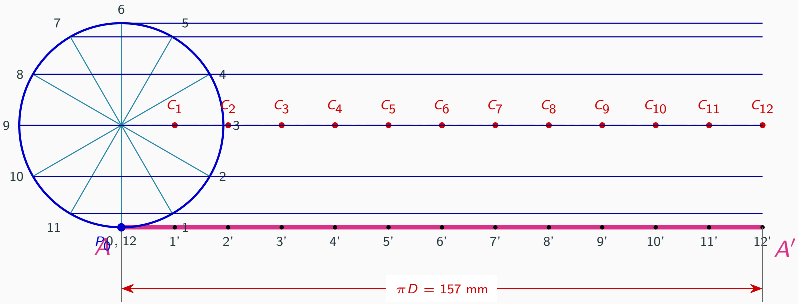


Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$

$C_1 \dots C_{12} \rightarrow$ Retained Centers
$P_1 \dots P_{12} \rightarrow$ Cutting Arcs/Points



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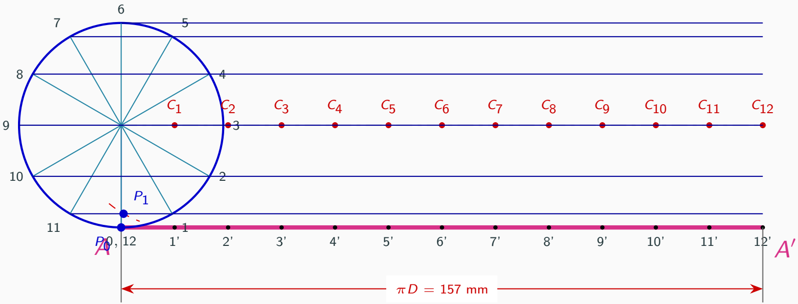

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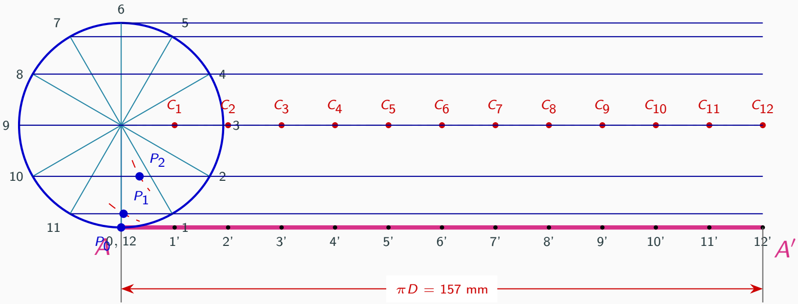

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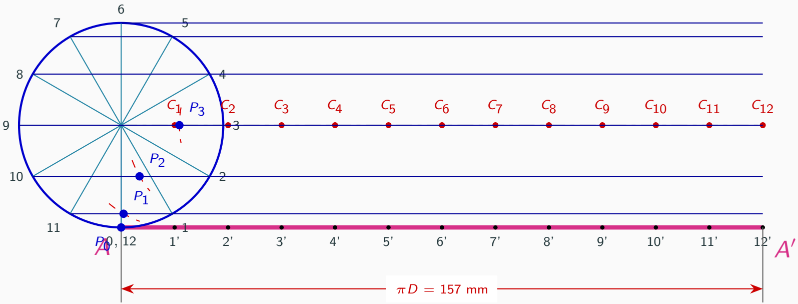

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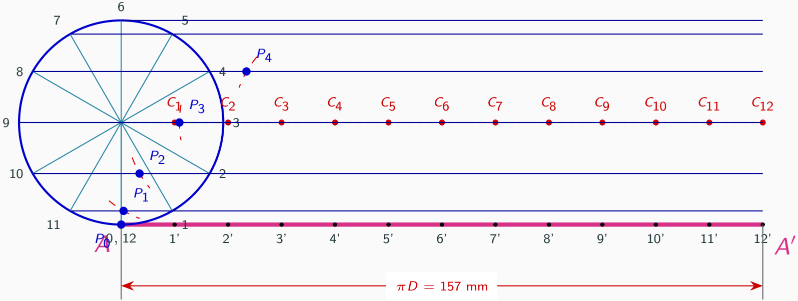

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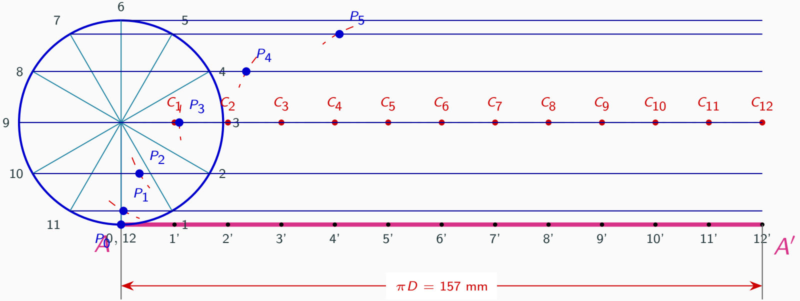

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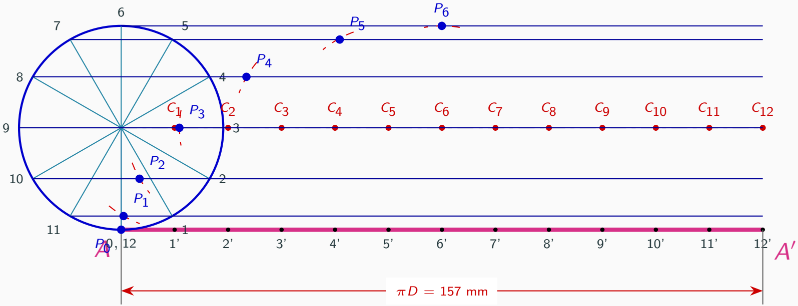

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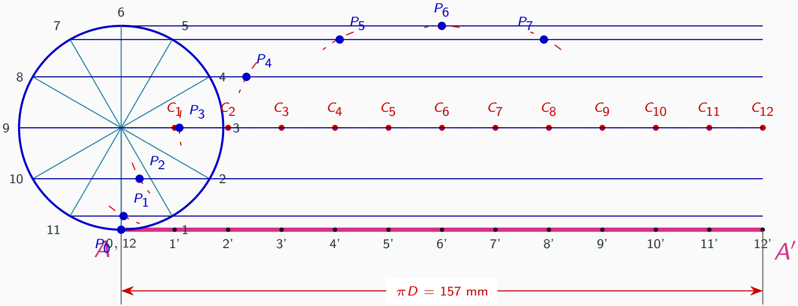

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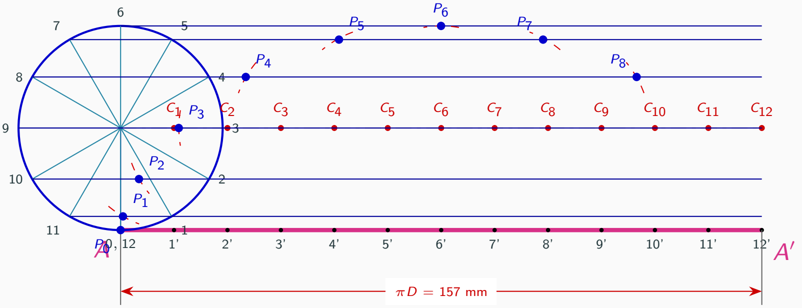

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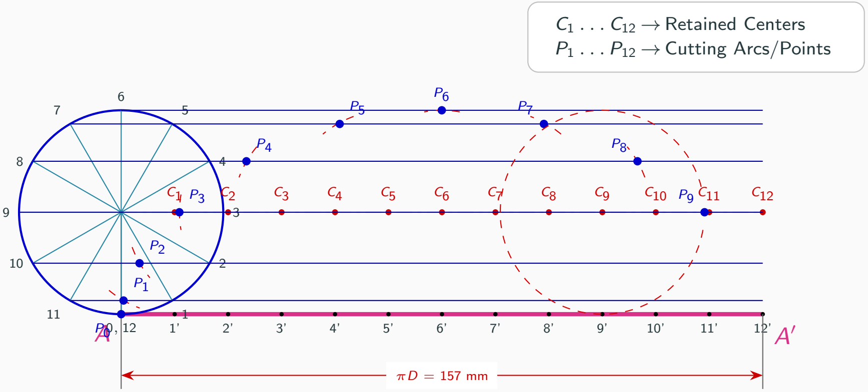

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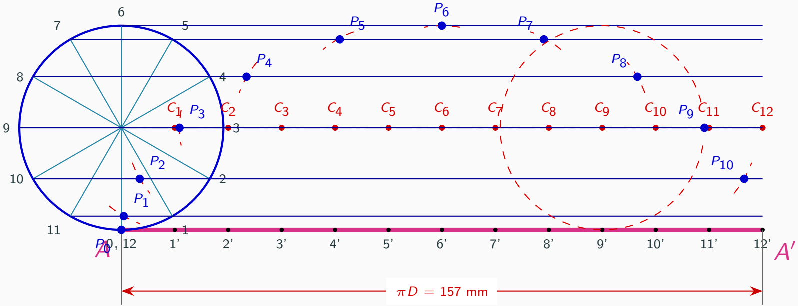

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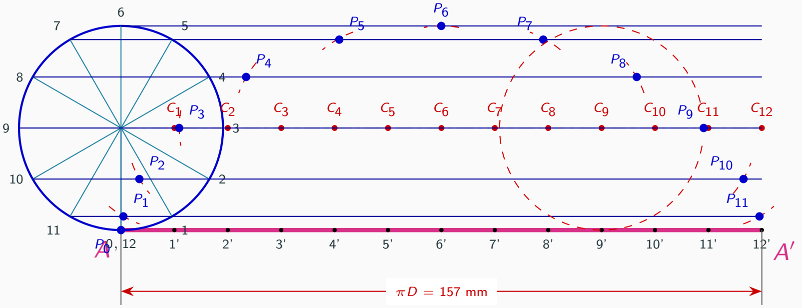

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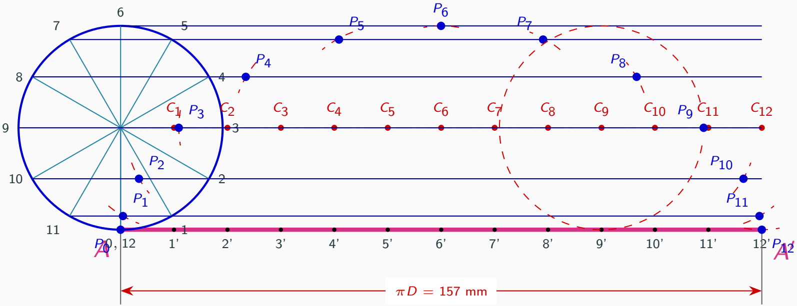
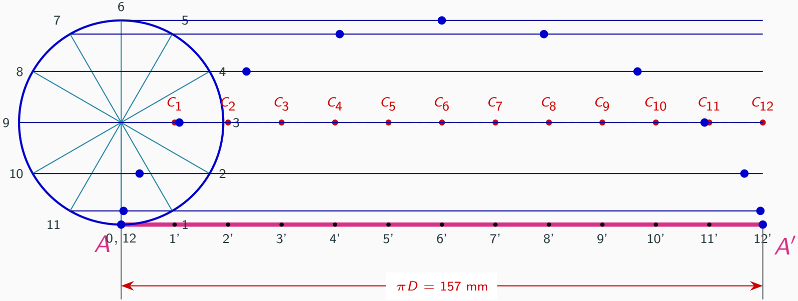


Step 7: Smoothly Join Points to Form Cycloid Curve

Curve $\longrightarrow$ Completed Profile
$P_0 \ldots P_{12} \longrightarrow$ Retained Points



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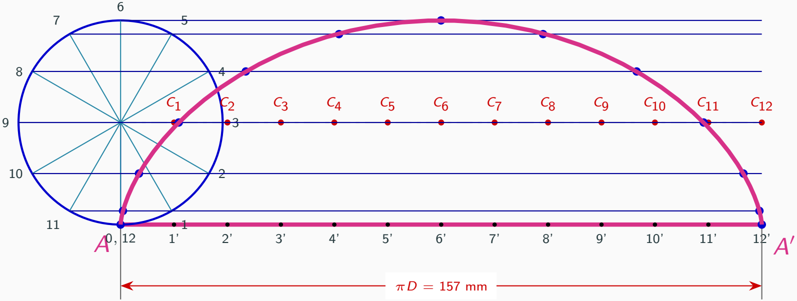
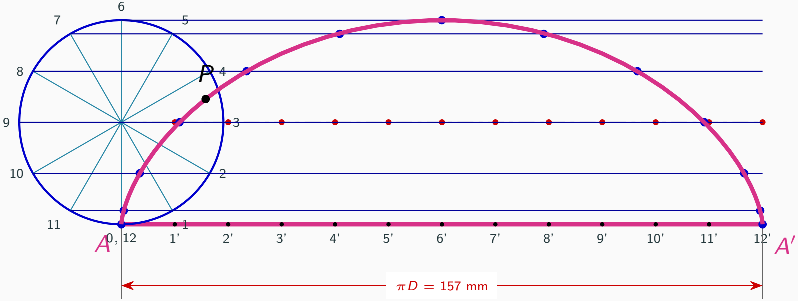


Step 8: Mark Point P & Locate Center C_p

$P \longrightarrow$ Target Point
$C_p \longrightarrow$ Corresponding Center



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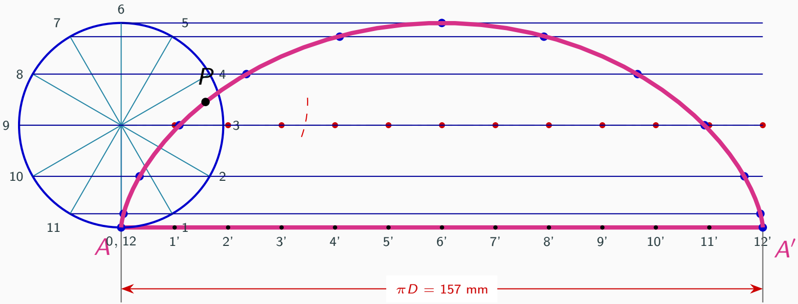

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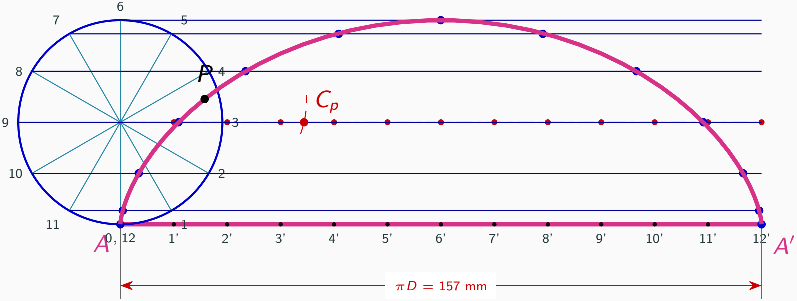

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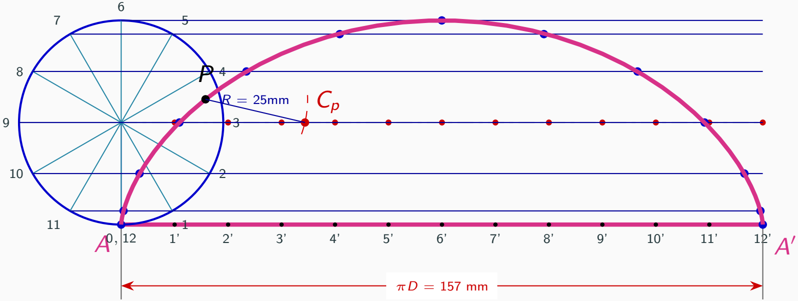
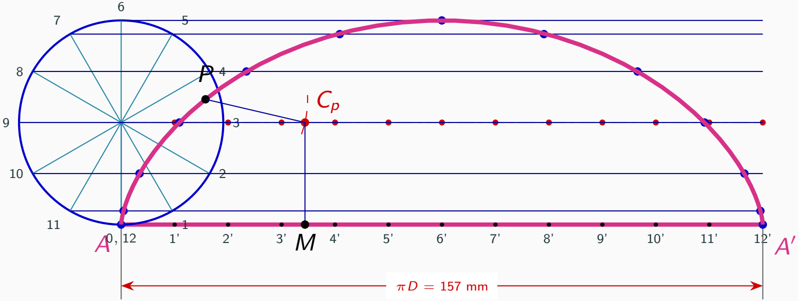


Step 9: Construct Normal Line N-N'

N-N' $\longrightarrow$ Normal Line
M $\longrightarrow$ Base Projection



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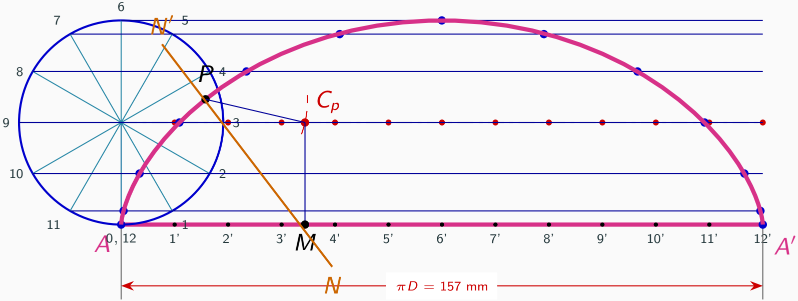
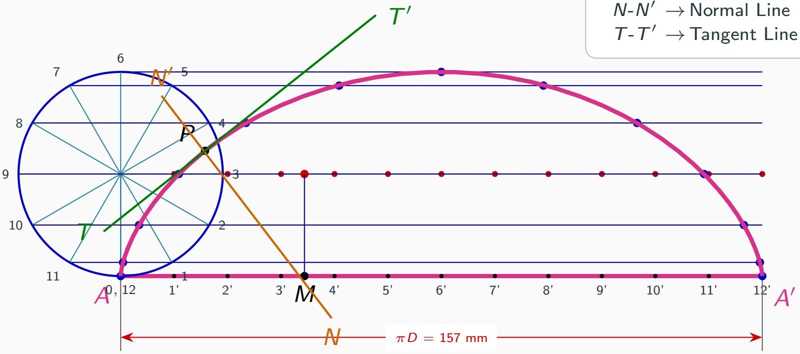


Step 10: Construct Tangent Line T-T' & Final Result

Curve $\to$ Completed Profile
N-N' $\to$ Normal Line
T-T' $\to$ Tangent Line



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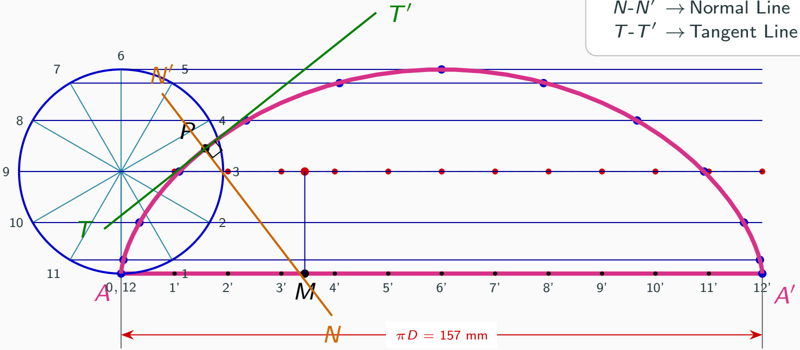

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